

When Shared Experience Becomes an Implementation Attractor: Supplementary Material

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1 Evidence Scope and Epistemic Labels

This supplement records the frozen protocol, deterministic audit rules, full trajectory summaries, and artifact map for the main paper. We distinguish three evidence classes throughout:

- **Frozen confirmatory experiment:** the 12-layer, three-arm Kimi-K2.6 `cfgpipe` run and its preregistered catastrophic-tail and local-admission decisions.
- **Retrospective replication audits:** two additional, independently executed cumulative self-report trajectories to which the exact-snapshot lock-in metric was applied after collection.
- **Prospective context-growth control:** a fresh six-layer run whose fixed-three-reference routing and two-consecutive-layer lock-in criterion were frozen before any new outcome was observed.
- **Prospective mitigation and task boundary:** a local-only diversity-router arm with its endpoint frozen before model calls, followed by a same-provider three-arm `log_query` probe frozen before all 27 new outcomes.

Layers within a run are coupled by shared history and are never treated as independent samples. Cross-provider and cross-task trajectories are reported separately rather than pooled.

2 Testbed and Frozen Main Protocol

2.1 Execution Topology

Each layer contains three MiniSWE workers that receive the same public task and edit separate repository workspaces. A barrier closes the complete layer before policy-selected output summaries can be delivered to the next layer. The hidden evaluator runs only after each worker finishes. External Core or all-test pass counts, failed-test names, and the test-collection hash are not inserted into self-report or local-admission recipient prompts.

The frozen all-test collection hash is `6ee2bfc068272a7fbc92599619db56b345cd93ecd3a321e791fa496a6a0f0615`. Both transfer arms receive the same three bootstrap sources, independently re-evaluated at 4/4 Core and 34/37 overall. Those scores are used only by the auditor and are withheld from recipients. The no-transfer arm receives no bootstrap source.

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Table 1: Frozen main-experiment configuration.

Field	Frozen value
Provider/model	ModelArts MaaS / Kimi-K2.6
Task	<code>cfgpipe</code> , checkpoint 1
External evaluator	4 Core tests; 37 total tests
Workers and depth	3 workers per layer; 12 layers
Agent budget	12 environment actions per worker
Sampling	Temperature 0.6; thinking disabled
Source truncation	12,000 characters per experience source
Transfer arms	Cumulative self-report; local admission
Control	No transfer
Local threshold	Claimed success and at least 3 successful commands

2.2 Transfer and Admission Semantics

2.3 Exact Worker Artifact and Admission Rules

Each worker artifact is generated from the following schema before the 12,000-character cap:

```
Worker <worker_id>, layer <layer>.  
Local verifier Core: <passed>/<total>;  
pytest exit <code>.  
Recent MiniSWE actions and reasoning:  
<trailing 6,000 characters from the  
final four assistant messages>  
Resulting source snapshot excerpt:  
<UTF-8 source files under the remaining  
artifact budget>
```

When self-report or local admission delivers an artifact, the numeric Core line is replaced by:

```
Agent completion report: accepted  
without an independent smoke check;  
external evaluation withheld.
```

The exact self-report expression is:

```
\b(pass|passes|passed|works|working|suc  
cess|successful|correct|verify|verified  
)\b
```

with case-insensitive matching over the concatenation of all assistant message content. The local gate additionally requires at least three successful environment commands. External evaluator outcomes are unavailable to both rules. The recursive manager receives the public task specification, previous playbook, and accepted source artifacts, and is instructed to retain constraints, implementation decisions, edge cases, failure-prevention rules, and validation procedures while resolving conflicts explicitly.

2.4 Frozen Decision Rules

Before new-arm outcomes, the main protocol required the cumulative self-report arm to produce at least two post-layer-6 0/37 outcomes, including at least one at layer 10 or later, to support recurrent complexity-tail risk. The no-transfer arm had to produce fewer such events. The local mechanism was supported if every observed 0/37 output with fewer than three successful commands was rejected and never appeared in a later recipient prompt.

The catastrophic-tail criterion was falsified: cumulative self-report produced no 0/37 outcomes in 36 worker-layer

evaluations. The local mechanism criterion passed: two zero-command 0/37 false completions were rejected and their artifact identifiers never appeared downstream. These decisions are separate from the later snapshot-lock-in audit.

3 Evaluation and Identity Audit

External score is the fraction of the frozen test collection passed. A valid catastrophic event is exactly 0/37 after a completed model query and a non-infrastructure evaluation. API, quota, rate-limit, or gateway failures are not scored as benchmark failures.

For implementation identity, the auditor recursively enumerates every file in the worker’s submitted checkpoint directory, sorts paths lexicographically, and updates SHA-256 with each UTF-8 relative path, a null byte, the complete file bytes, and a second null byte. Three-worker exact lock-in at a layer requires three valid snapshots with one shared directory hash. Behavioral identity is separately measured from the set of failed test names. Exact snapshot identity is therefore a strong repository-state equality test, but it does not establish semantic equivalence beyond the evaluator.

The new prospective mitigation and boundary additionally freeze a *primary implementation* identity: SHA-256 of `cfgpipe.py` or `logql.py`. The diversity router independently hashes local source files with conventional implementation suffixes while excluding test files, caches, and evaluator outputs. This distinction matters in the bounded control: layers 4 and 6 have two full-workspace hashes because one worker leaves an auxiliary `ws.txt` or test file, but all workers share the same `cfgpipe.py` from layers 2–6.

The transfer-mechanism audit also reports exact prior-primary match, normalized source-line Jaccard, function-name Jaccard, AST-node-count signature Jaccard, string-literal Jaccard, and the fraction of recipient source lines available verbatim in at least one delivered snapshot excerpt. These metrics measure alignment and prompt availability; they do not distinguish literal copying from deterministic regeneration.

Every interpreted trajectory is checked for:

- the frozen Core count, all-test count, and collection hash;
- at least one completed model step and no terminal model error;
- absence of hidden scores, failed-test names, and collection hashes in recipient prompts;
- absence of assistant actions that access benchmark catalogs, hidden tests, or external-evaluation paths; and
- stale evaluator files, which are logged but not substituted for the current validated evaluation.

4 Independent Lock-In Replication Audit

Table 3 reports all trajectories in the deterministic audit. “First lock-in” is the first of at least two consecutive layers with one exact snapshot across three valid workers. The earliest Coding self-report run exposed bootstrap Core scores and is retained only as a protocol-drift observation; it is not counted as an independent replication.

Table 2: Operational policy definitions. “External” refers to the hidden evaluator and is unavailable to recipients unless a mechanism probe is explicitly labeled as Core-gated.

Policy	State delivered to each next-layer worker	Admission signal	Main use
No transfer	Empty state	None	Same-provider control
Cumulative self-report	Every accepted bounded summary to date	Trajectory claims successful completion	Main treatment and retrospective replications
Bounded recent	Three most recent accepted summaries; bootstrap is replaced after a complete generated layer	Same self-report signal	Prospective context-growth control
Bounded diverse	Latest three locally distinct implementation identities across bootstrap and completed layers	Same self-report signal plus local source hash	Prospective mitigation and task boundary
Local admission	Every accepted bounded summary to date	Claimed success and successful commands ≥ 3	Prospective rejection mechanism
Recursive manager	One manager-revised playbook	Self-report or external Core gate, depending on probe	Compression and task-boundary probes

Table 3: Trajectory-level lock-in audit. Terminal score is the three-worker mean normalized all-test score.

Run	Role	Depth	First lock-in	Terminal snapshots	Terminal score
Kimi Coding no transfer	Control	5	—	3	0.919
Kimi Coding self-report r1	Protocol drift only	8	3	1	0.613
Kimi Coding self-report r2	Retrospective replication	10	3	1	0.919
Kimi MaaS self-report r1	Retrospective replication	12	4	1	0.613
Kimi MaaS no transfer	Control	12	—	3	0.874
Kimi MaaS self-report r2	Frozen main treatment	12	5	1	0.892
Kimi MaaS bounded recent r1	Prospective mechanism control	6	2	2	0.919
log_query no transfer	Second-task control	5	—	3	0.938
log_query verified cumulative	Second-task boundary	5	—	3	0.963

The independent Coding r2 trajectory has exact three-worker snapshot convergence at layers 3–10. The independent MaaS r1 trajectory converges at layers 4–9, has an intermittent no-snapshot 0/37 event at layer 10, reconverges at layer 11, and again has a no-snapshot event at layer 12. Thus lock-in is a recurring attractor in these trajectories, not evidence of permanent irreversibility. Neither matched no-transfer control has two consecutive exact-convergence layers.

5 Prospective Fixed-Reference Control

The bounded-control configuration and outcome rule were frozen on 2026-07-15 before new outcomes. Every worker receives exactly the three most recent self-reported outputs. Once a complete generated layer is available, those outputs replace the bootstrap sources; the reference count remains three. The frozen success criterion was at least two consecutive exact-lock-in layers by layer 6.

The frozen full-workspace criterion passes at layers 2–3. The later apparent re-diversification is auxiliary-file churn: the primary implementation remains identical from layers 2–6. This control excludes cumulative reference-count growth as a necessary condition in the tested model-task pair. It does not exclude transferred context or token load as contributors because three summaries are still added relative to no transfer.

Table 4: Complete bounded-control trajectory. “Workspace” hashes every file; “primary” hashes only `cfgpipe.py`. All workers pass 34/37 and 4/4 Core at every layer. Input ratio uses the matched same-provider no-transfer layer.

Layer	Refs	Workspace	Primary	Primary exact	Input ratio
1	3	3	3	No	2.80
2	3	1	1	Yes	1.26
3	3	1	1	Yes	1.60
4	3	2	1	Yes	0.99
5	3	1	1	Yes	1.30
6	3	2	1	Yes	1.14

6 Experience Alignment and Prospective Mitigation

The worker artifacts are code-rich rather than abstract memories. Median output artifact size is 10,479 characters in the cumulative main arm and 10,611 in the bounded arm; the median trajectory component reaches its 6,001-character budget, while the median snapshot excerpt contributes 4,338 and 4,470 characters, respectively. In cumulative transfer, received experience grows to 367,725 characters and per-worker input reaches 703,436 tokens. The bounded arm keeps received experience between 25,934 and 32,082 characters.

The `cfgpipe` mitigation protocol freezes the local-only diversity router, its implementation hashes, and the following primary endpoint before its model calls: the sum of unique

Table 5: Mechanism audit after layer 1. “Exact” is the fraction of recipients matching at least one prior primary implementation. Line metrics take the best available prior source per recipient.

Run	Exact	Source-line J	Summary-line recall	Median received chars
Cumulative main	0.879	0.965	0.976	193,858
Bounded recent	1.000	1.000	1.000	31,706

Table 6: Prospective cfgpipe diversity-routing result. AUC is the sum of unique primary implementations over layers 2–6.

Arm	AUC	Diverse layers	Lock-in	Mean tests	Input tokens
No transfer	15	5	–	30.0/37	994,496
Bounded recent	5	0	L2	34.0/37	1,399,159
Bounded diverse	6	1	L3	34.0/37	1,507,917

primary implementations over layers 2–6. The already frozen no-transfer and bounded-recent outcomes were known and explicitly reused; only the new mitigation arm is prospective.

The frozen support threshold is exceeded, but the stronger criterion is not: local diversity routing retains two implementations only at layer 2 and then locks through layer 6. The result is classified as partial support, a transient one-layer delay with no score change.

7 Prospective Same-Provider Task Boundary

The `log_query` protocol freezes three layers, three MaaS Kimi-K2.6 workers, identical source and action limits, and three arms before all 27 new outcomes. The two transfer arms share three restored bootstrap sources, each 10/10 Core, self-report accepted, and locally implementation-distinct.

All target, model-query, blinding, reference-count, and prohibited-access gates pass. Bounded recent retains three implementations at both post-transfer layers, so the frozen cfgpipe lock-in criterion does not reproduce within this horizon. The diversity-mitigation comparison is ceiling-limited and not identifiable: bounded recent already preserves the maximum diversity. This boundary does not exclude later lock-in at greater depth.

8 Admission Threshold Audit

The three-command arm was prospectively frozen, while the sweep in Table 8 is a retrospective replay over its 36 completed outputs. High quality is defined here as at least 33/37. The sweep measures a sample-specific trade-off and does not identify a universal optimum.

9 Cross-Model Tails and Mechanism Boundaries

Across five cumulative self-report runs, DeepSeek-V4-Flash produces 2/30 valid zero-score outcomes, GLM-5 produces 0/30, and Kimi-K2.6 produces 1/24, 0/30, and 0/36 across three endpoint/run combinations. All observed zero events are isolated or terminal. Provider and model runs are not treated as exchangeable replications.

Recursive manager synthesis is also directionally conditional. On `cfgpipe`, a Core-gated Kimi manager reaches

Table 7: Prospective `log_query` boundary. Diversity AUC is the sum of unique primary implementations over layers 2–3; the maximum is 6.

Arm	AUC	L2/L3 unique	Mean Core	Mean tests	Input tokens
No transfer	6	3/3	7.89/10	108.44/134	544,396
Bounded recent	6	3/3	9.67/10	127.22/134	1,279,735
Bounded diverse	6	3/3	9.67/10	130.11/134	1,250,575

Table 8: Retrospective local-gate threshold replay.

Threshold	Accepted	0/37 rejected	HQ retained	HQ rate
0	36	0/2	34/34	1.000
1	34	2/2	34/34	1.000
2	34	2/2	34/34	1.000
3	29	2/2	29/34	0.853
4	22	2/2	22/34	0.647
5	13	2/2	13/34	0.382
6	5	2/2	5/34	0.147

exact terminal snapshot lock-in at 32/37, while no transfer retains three snapshots at 34/37. A DeepSeek stress probe compresses a mixed-quality bootstrap into one shared failure signature across distinct implementations. In contrast, recursive synthesis improves terminal normalized score by 0.047 on `log_query` and 0.072 on `xjq`. The separate strictly Core-gated cumulative `log_query` boundary ends with three distinct snapshots and scores 0.963 versus 0.938 for no transfer. These probes establish task- and admission-conditional direction, not a general harmful effect. The new same-provider three-arm boundary strengthens this limit: both self-report transfer policies retain three primary implementations at every measured layer while raising mean all-test count relative to no transfer.

10 Reproducibility Artifacts

The following repository-relative artifacts form the evidence chain:

- main frozen protocol: `benchmarks/scbench/kimi_maas_cfgpipe_cp1_depth12_three_arm_confirmation_protocol_20260715.json`;
- main audit: `benchmarks/scbench/kimi_maas_cfgpipe_cp1_depth12_three_arm_confirmation_audit_20260715.json`;
- bounded configuration and freeze sidecar: `benchmarks/scbench/kimi_maas_cfgpipe_cp1_bounded_context_depth6_config.json` and `benchmarks/scbench/kimi_maas_cfgpipe_cp1_bounded_context_depth6_config.freeze.json`;
- experience schema and transfer-mechanism audit: `paper_draft/EXPERIENCE_ARTIFACT_SCHEMA_20260715.md` and `benchmarks/scbench/scbench_transfer_mechanism_audit_20260715.json`;
- prospective mitigation protocol and audit: `benchmarks/scbench/kimi_maas_cfgpipe_`

cpl_mitigation_three_arm_protocol_20260715.json and benchmarks/scbench/kimi_maas_cfgpipe_cpl_mitigation_three_arm_audit_20260716.json;

- **prospective second-task config and audit:** benchmarks/scbench/kimi_maas_log_query_cpl_three_arm_boundary_depth3_config.json and benchmarks/scbench/kimi_maas_log_query_cpl_three_arm_boundary_depth3_audit_20260716.json;
- **deterministic cross-trajectory audit:** scripts/audit_scbench_lockin_replications.py;
- **audit outputs:** benchmarks/scbench/kimi_cfgpipe_lockin_replication_audit_20260715.json and benchmarks/scbench/kimi_cfgpipe_lockin_replication_audit_20260715.md;
- **layer-level source data:** paper_draft/generated/moa_paper/lockin_replication_layers.csv; and
- **bounded raw run:** runs_scbench_moa_kimi_maas_cfgpipe_cpl_bounded_context_depth6_r1_20260715.

The bounded configuration SHA-256 is 08de65c5f3ff5711d99d477cc4db9f4bdf2bc8f271e4d21007f2b3440a0a7f4d. The three-arm protocol SHA-256 at this evidence freeze is 599a3c4b6d99caaa4d96199a413947303ddda6e1b80de560bac56ebda4497a6c. The prospective mitigation protocol SHA-256 is bd8c9cc738b1bc01c808edd8e46fd72e1af1b6e45c0df563986df82de5f1958c, and the prospective log_query configuration SHA-256 is ee b7664d11acc22376a0291a9670d5cd7c9a779347e52adb3de3a01d10a9933.

To reproduce the deterministic audit and figures from existing run artifacts:

```
pythonscripts/audit_scbench_lockin_replications.py
pythonscripts/audit_scbench_transfer_mechanism.py
pythonscripts/audit_scbench_mitigation_three_arm.py
pythonscripts/audit_scbench_log_query_boundary.py
pythonpaper_draft/make_moa_figures.py
```

11 Computing Environment and Claim Boundary

The final audit and paper build use Python 3.12.12 and TeX Live 2026. Model calls use the provider endpoints recorded in each frozen JSON configuration; API credentials are supplied only through environment variables and are not stored in run manifests or paper artifacts.

The evidence supports a conditional, multi-layer implementation attractor under blinded experience transfer. Local identity-aware routing delays this attractor by one layer on cfgpipe but does not prevent it; on the prospective log_query boundary, transfer improves quality without

reducing measured diversity. The evidence does not show that context length is irrelevant, that lock-in is permanent, that sharing is generally harmful, or that a command-count threshold is universally optimal. Exact hashes and failed-test signatures are complementary audited measurements bounded by the submitted repository state and frozen test suite.